

6.1.1 INCREMENTED DATE

ALGORITHM

Step 1: Start

Step 2: Input day, month, year

Step 3: If $\text{year} \leq 0 \rightarrow$ Print "Invalid Date" and Stop

Step 4: If $\text{month} < 1$ OR $\text{month} > 12 \rightarrow$ Print "Invalid Date" and Stop

Step 5: Determine number of days in the month:

If $\text{month} = 1, 3, 5, 7, 8, 10, 12 \rightarrow \text{days} = 31$

If $\text{month} = 4, 6, 9, 11 \rightarrow \text{days} = 30$

If $\text{month} = 2$:

If leap year $\rightarrow \text{days} = 29$

Else $\rightarrow \text{days} = 28$

(Leap year condition:

$\text{year} \% 400 == 0$ OR

$(\text{year} \% 4 == 0 \text{ AND } \text{year} \% 100 \neq 0)$)

Step 6: If $\text{day} < 1$ OR $\text{day} > \text{days} \rightarrow$ Print "Invalid Date" and Stop

Step 7: If $\text{day} < \text{days}$

$\text{day} = \text{day} + 1$

Step 8: Else ($\text{day} == \text{days}$)

$\text{day} = 1$

If $\text{month} == 12$

$\text{month} = 1$

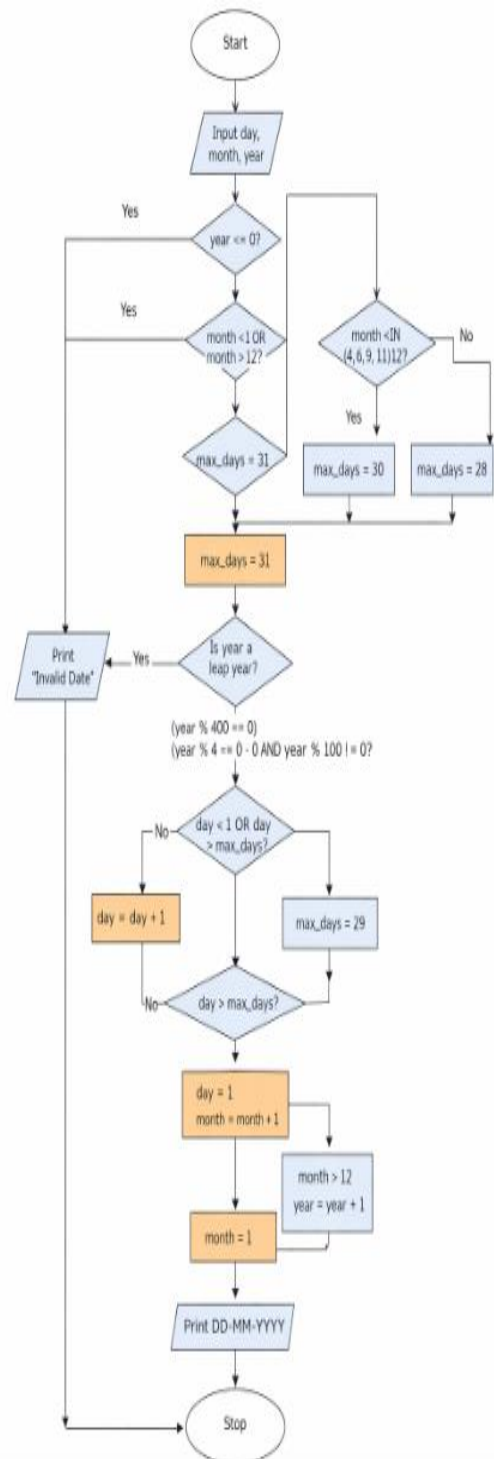
$\text{year} = \text{year} + 1$

Else

$\text{month} = \text{month} + 1$

Step 9: Print date in DD-MM-YYYY format

Step 10: Stop



CODETANTRA:-

CODETANTRA

Home

aryan.kamdi.batch2025@sitnagpur.siu.edu.inSupportLogout

6.1.1. Incremented Date08:52

Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".

Date Validation Rules:

1. Valid Month: 1 to 12

2. Valid Day: Depends on the month and year

- January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
- April (4), June (6), September (9), November (11): 1 to 30 days
- February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year

3. Valid Year: Any positive integer greater than zero.

Date Increment Rules:

If the date is valid, increment it by one day:

- If it's the last day of the month, move to the 1st day of the next month
- If it's December 31st, move to January 1st of the next year

Sample Test Cases

nextDate.py

```
1 day = int(input())
2 month = int(input())
3 year = int(input())
4
5 def is_leap_year(year):
6     if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
```

Explorer

Debugger

Average time0.045 sMaximum time0.066 s

5 out of 5 shown test case(s) passed5 out of 5 hidden test case(s) passed

Test case 153 msDebug

Expected output

Actual output

15	15
3	3
2024	2024
16-03-2024	16-03-2024

Terminal

Test cases

Test case 258 ms

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